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Enhancing Stock Price Prediction with MACD and EMA Features Using LSTM Algorithm

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***Abstract*: This research paper proposes the application of the Moving Average Convergence Divergence (MACD) and Exponential Moving Average (EMA) indicators in combination with the Long Short-Term Memory (LSTM) algorithm for stock price prediction. This study aims to enhance the accuracy and robustness of stock market predictions by using the strengths of technical indicators and deep learning methodologies. The LSTM algorithm, known for its ability to model long-term time series data, is used in this research. In this paper, we considered various evaluation metrics such as RMSE, R2 score, accuracy, precision, recall, and F1 score. Real-world historical stock price data, i.e., RELIANCE, KOTAK BANK, INFOSYS, ASIAN PAINT, HINDUSTAN UNILEVER, HDFC BANK, VEDANTA, and ICICI BANK are utilized to validate the proposed model, ensuring practical relevance and applicability. The experimental results demonstrate that the integration of MACD and different variants of EMA (26-day EMA, 100-day EMA, and 200-day EMA) features with LSTM significantly improves stock price prediction accuracy. The results show that the accuracy is improved by using the indicators MACD and EMA, as well as basic features such as open, high, low, and close (OHLC) features. Good precision values indicate that the proposed model captures positive events effectively. Good recall values indicate that the model can predict true positive values correctly, and the F1-score is also good, which indicates that the model has a good combination for effectively capturing positive and negative instances. The RMSE is computed on a testing dataset to assess the model's performance. A lower RMSE indicates better performance. The R2 score was 0.75 < R2 < 0.95, which indicated a strong model.**

Keywords - Stock Market Prediction, Machine Learning (ML), Deep Learning (DL), Moving Average Convergence Divergence (MACD), Exponential Moving Average (EMA), Long Short Term Memory (LSTM)

I. INTRODUCTION:

Due to the dynamic and complex nature of financial markets, effective tools and methodologies are needed to predict stock prices effectively. Investors, traders, and financial analysts are continuously exploring innovative approaches to improve decision-making related to the stock market. Traditional technical analysis plays an important role in market forecasting [1]. However, the advent of machine learning techniques, particularly deep learning algorithms, has opened new directions for enhancing prediction accuracy and capturing market patterns [2]. This research paper proposes the fusion of classical technical analysis indicators, specifically the Moving Average Convergence Divergence (MACD) and Exponential Moving Average (EMA), with the advanced Long Short-Term Memory (LSTM) algorithm. Our objective is to highlight the strengths of both technical indicators and deep learning methodologies to improve the accuracy and reliability of stock price predictions. The MACD and EMA are well-established indicators widely used for trend identification, momentum analysis, and signaling potential reversal points in financial markets [3]. On the other hand, LSTM, a variant of recurrent neural networks (RNNs), is good at capturing difficult patterns and dependencies in sequential time series data. LSTM is well suited for modeling the dynamic nature of stock prices [4]. The integration of MACD and different versions of EMA features with LSTM produces better results. By doing so, we seek to develop a model that not only leverages the interpretability of traditional indicators but also adapts to the complex and evolving patterns inherent in financial time series. This research contributes to the evolving area of stock price prediction by proposing a comprehensive approach that bridges traditional and latest technologies [5]. The results from this study provide direction for practitioners, researchers, and stakeholders navigating the nonstationary and complex world of financial markets, providing an approach based on the fusion of technical analysis and deep learning for stock price forecasting.

II. LITERATURE REVIEW:

Mehar Vijh et al. introduced a study in 2020. In this study, a machine learning approach is used to predict stock prices efficiently. The ANN algorithm is used along with OHLC features. The model shows good performance in terms of RMSE and MAPE [6]. A. I. Al-Alawi et al. reviewed the analysis of stock prediction techniques using machine learning. The findings showed that the LSTM and gated recurrent unit techniques provided the best results [7]. Using a deep learning approach, Hu, Z., et al. performed a survey of forex and stock price prediction. He focused on the LSTM, CNN, RNN, and HAN algorithms. This study highlights the importance of deep learning in stock and forex price forecasting [8]. In 2023 Burak Gülmez Introduced a study in which the LSTM algorithm was used to perform stock price prediction; the hyperparameter tuning of LSTM was performed using the Artificial Rabbits Optimization algorithm (ARO) algorithm. The results indicate that LSTM-ARO performs well in terms of MAE, MAPE, and R2 score evaluation [9]. S. Chen and C. Zhou introduced research in 2021 in which genetic algorithms were used for feature selection purposes in the LSTM algorithm. The GA-LSTM hybrid approach outperforms other time series forecasting models. The CSI 300 stock dataset is used for this study [10]. Yufeng Chen et al. (2022) provided a combination of K-means and LSTM algorithms for this study. China's commercial bank shares were selected as the dataset. K-means is used to construct clusters of similar trend stocks, after which,

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the LSTM algorithm is applied to that cluster to refine the results. The results showed that the hybrid models outperform the other models in terms of the MAE, MSE, and R-square [11]. In 2022, Y. Zhou and J. Zheng introduced a study in which the k-means algorithm was used, which considered different factors that affect the stock market. The results showed that the model provides high accuracy in predicting stock prices [12]. K. R. Reddy et al. introduced a study in 2022 in which the RNN algorithm was used for stock price forecasting. The experiments were performed with different epochs [13].

A. Research Gap:

Existing research has limited the integration of traditional features such as open, high, low, close (OHLC) features with deep learning models such as RNN and Long Short-Term Memory (LSTM). Researchers have focused only on open, high, low, and close (OHLC) values as features for stock price prediction; we have focused on more valuable indicators, such as the MACD and EMA, which act as additional advanced features to the model. In this research, we focused on the combination of MACD and different versions of EMA, i.e., 26-day EMA, 100-day EMA, and 200-day EMA, along with basic OHLC features.

III. METHODOLOGY:

A. System Architecture:

Fig 1. Shows the system architecture selected to implement the model for stock price and trend prediction purposes.

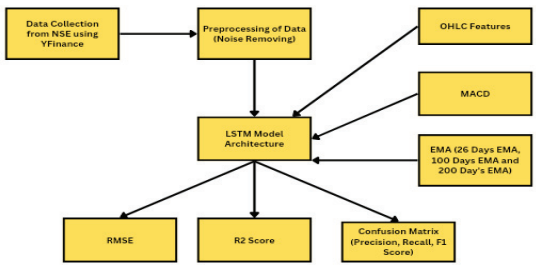

Fig. 1. System Architecture

a. **Data Collection:** Gather historical stock price data from NSE, India, using yfinance for a representative set of financial instruments [14]. Relevant features such as open, high, low, and close prices are included. The data range from January 1, 2020, to December 30, 2023.

b. **Preprocessing:** Clean and preprocess the collected data, and handle missing values and outliers. Normalize or scale the data to ensure uniformity across features.

c. **Feature Engineering:** Compute the MACD and EMA (i.e. 26 Days EMA, 100 Days EMA and 200 Days EMA) indicators from the historical stock price data. Consider additional relevant features that might enhance the prediction performance.

d. **Model Architecture:** The LSTM model architecture is designed considering the input features, number of layers, units per layer, and activation functions.

e. **Data Splitting:** The dataset was split into training and test sets; 80% of the training data and 20% of the testing data were used. A representative distribution of the data in each set was ensured.

f. **Model Training:** The LSTM model was trained on the training dataset (80%). The testing dataset (20%) was utilized to fine-tune the hyperparameters and prevent overfitting.

g. **Evaluation Metrics:** Appropriate evaluation metrics such as accuracy, precision, recall, F1 score, R2 score, and root mean squared error (RMSE) are chosen for regression tasks. The model's performance was evaluated on the test set.

B. LSTM Algorithm:

Fig 2. Represent the structure of the LSTM cells. The LSTM can store important data and forget unnecessary data in a time series sequence.

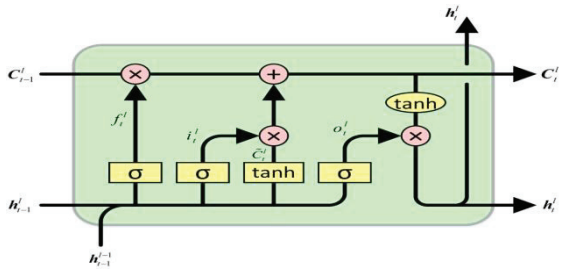

Fig. 2. LSTM cell structure

C_{t-1}^l represents the cell's previous state and C_t^l indicates the current state which we want to update. The LSTM determines which data we want to discard and which we want to preserve for the next state. The decision to forget is represented by 0 and the decision to remain is represented by 1. These operations are performed using the sigmoid activation function σ. The forgot operation is represented using equation (i)

$$f_t^l = \sigma(W_f \cdot [h_{t-1}^l, h_{t-1}^{l-1}] + b_f) \tag{1}$$

The new input is taken into the cell and the formula for input gate is represented using equation (ii)

$$i_t^l = \sigma(W_i \cdot [h_{t-1}^l, h_{t-1}^{l-1}] + b_i) \tag{2}$$

The tanh activation layer is used to select the value that we want to insert in the state. Tanh generates values between -1 and +1 and multiplies it with the output of the sigmoid layer. The output gate is represented as equation no (iii).

$$o_t^l = \sigma(W_o \cdot [h_{t-1}^l, h_{t-1}^{l-1}] + b_o) \tag{3}$$

h_{t-1}^l and h_t^l are responsible for providing new input data to the cell state. LSTM is better than RNN because RNN faces the "Vanishing Gradient" problem which means that they cannot handle long time series sequences [15].

C. MACD:

The moving average convergence divergence is the leading indicator used for the technical analysis of stocks. It contains two lines. The first is the fast line, which is the difference between the 12-day EMA and 26-day EMA. The second is the slow line, which is the 9-day EMA. The slow line is also called a signal line, which gives buy and sell signals. Fig 3. Indicate

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the simple structure of the MACD of the sample stock, i.e., RELIANCE. The MACD indicator is used as a feature in our model, along with OHLC features.

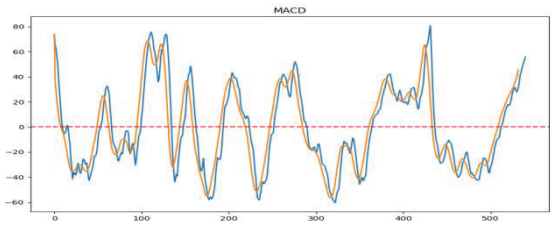


Fig. 3. MACD structure for RELIANCE stock.

D. EMA:

The EMA is the technical indicator that is used for the technical analysis of stocks. It is a weighted simple moving average line. The crossover of this line indicates the buy or sell signal for stock. For this study, we selected three EMAs, i.e., 26 days EMA, 100 days EMA, and 200 days EMA. The different variants of EMA will yield better results. The structure of the EMA is shown in Fig. 4. It is the representation of EMA for sample stock, i.e., RELIANCE.

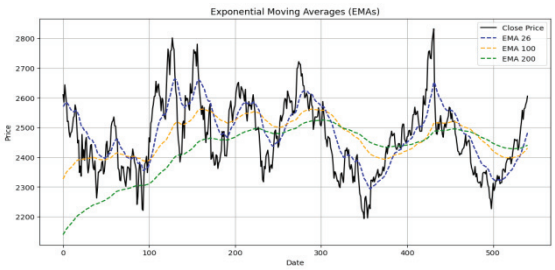


Fig. 4. EMA structure for RELIANCE stock

IV. Results and Discussion:

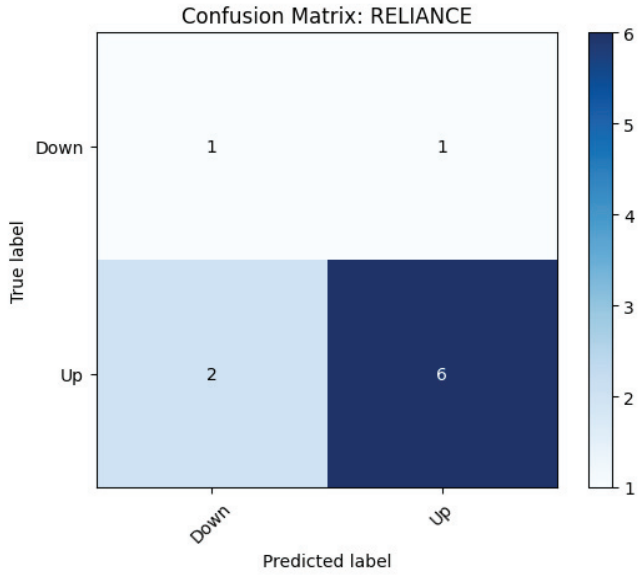
For this study, we selected eight fundamentally good stocks. When we apply prominent indicators such as the MACD and EMA (i.e., 26-day EMA, 100-day EMA, and 200-day EMA), the RMSE and R2-Score are obtained, as shown in Table 1. The training data and testing data values are measured separately, as shown in the table. The R2-scores for Reliance, Kotak Bank, Infosys, Asian Paints, Hindustan Unilever, HDFC BANK, VEDANTA, and ICICI Bank are $0.75 < R2 < 0.95$, which indicates a strong model. The RMSE decreases when we apply 20% of the testing data because the model is efficiently trained by using 80% of the training data and becomes familiar with the behavior of the stock.

TABLE I. RMSE and R2-Sore values of the selected stock forecasts.

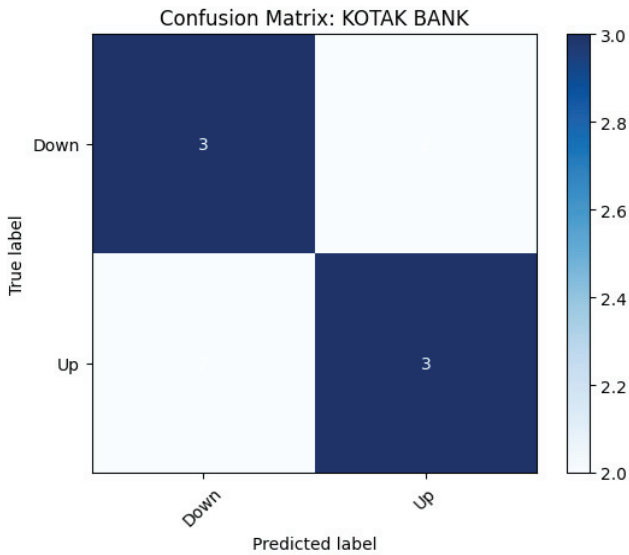
Name of Stock	RMSE		R2-Score	
	TRAINING	TESTING	TRAINING	TESTING
RELIANCE	53.98	33.51	0.82	0.86
KOTAK BANK	39.93	25.92	0.84	0.85
INFOSYS	37.26	41.87	0.95	0.75
ASIAN PAINTS	76.17	50.71	0.93	0.83
HINDUSTAN UNILEVER	47.99	34.37	0.94	0.76
HDFC BANK	32.99	30.79	0.89	0.78
VEDANTA	11.19	9.79	0.97	0.77
ICICI BANK	19.53	17.76	0.97	0.81

A. Confusion matrix:

A confusion matrix was used to evaluate the classification. Our developed model can predict the next day's trend in the stock. The trends are classified into two categories: uptrends (class 1) and downtrends (class 0). The stock trend readings were taken for 10 days to evaluate the model. Fig. 5 represents the confusion matrix for all the selected stocks.



a) RELIANCE



b) KOTAK BANK

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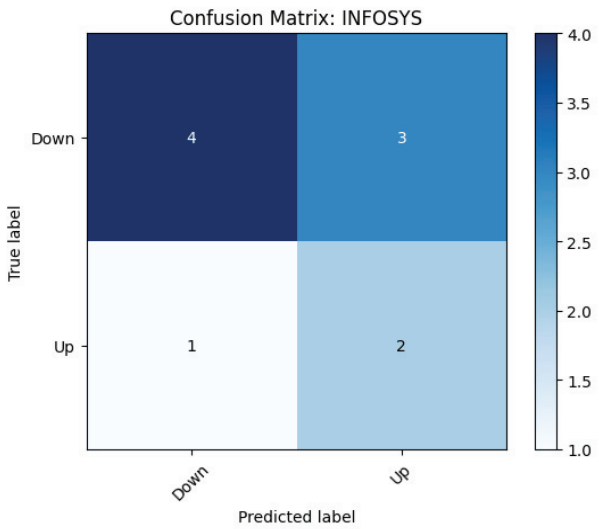
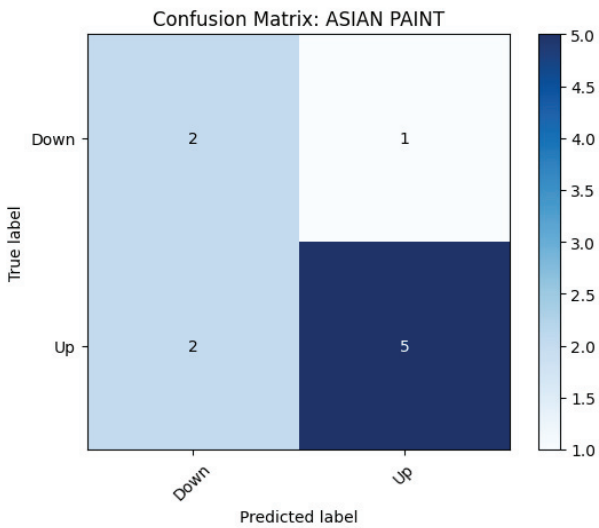
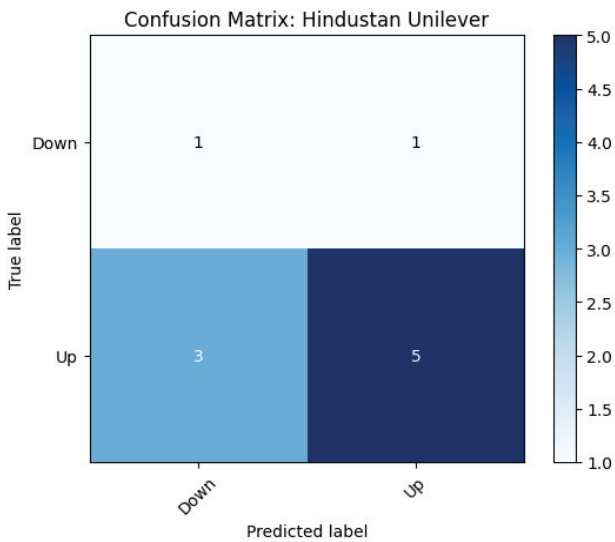


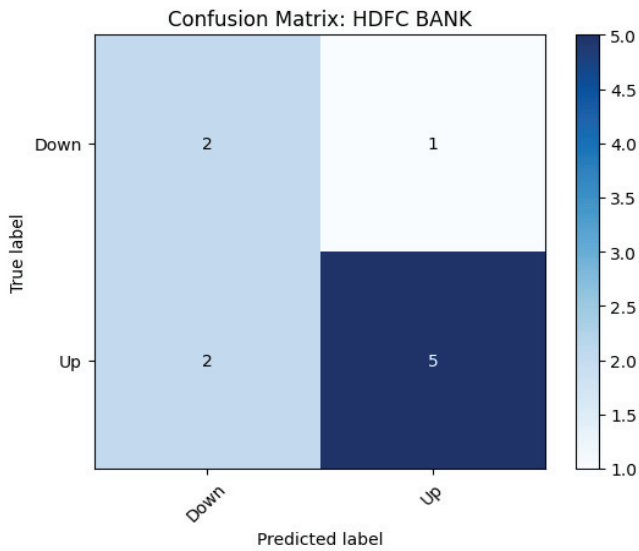
c) INFOSYS



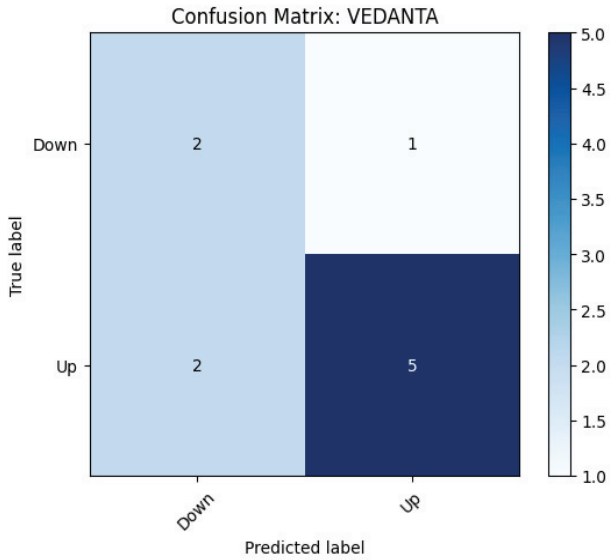
d) INFOSYS



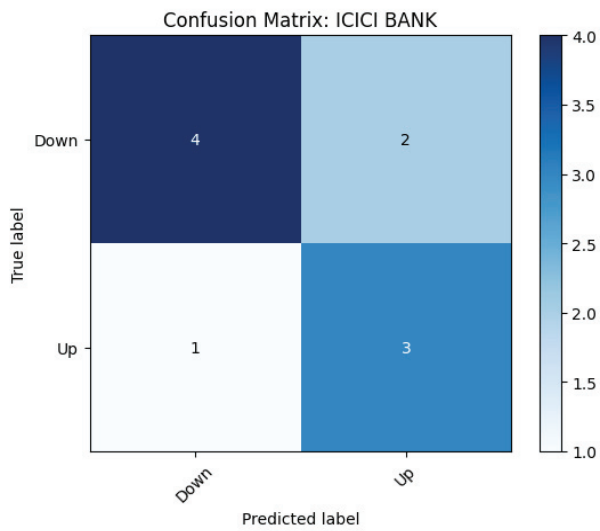
e) HINDUSTAN UNILEVER



f) HDFC BANK



g) VEDANTA



h) ICICI BANK

Fig. 5. Confusion matrix of RELIANCE, KOTAK BANK, INFOSYS, ASIAN PAINTS, HINDUSTAN UNILEVER, HDFC BANK, VEDANTA and ICICI BANK stocks.

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Table 2 clearly shows that the accuracy achieved reaches 60% in the Kotak Bank, Infosys, and Hindustan Unilever datasets. The accuracy of Asian Paint, Reliance, HDFC Bank, Vedanta, and ICICI Bank reaches 70%. The precision values are good for Reliance (0.86), Asian paints (0.83), Hindustan Unilever (0.83), HDFC Bank (0.83), and VEDANTA (0.83) which indicate that the proposed model can predict positive predictions effectively. The recall values are good for Reliance (0.75), Asian Paints (0.71), HDFC Bank (0.71), VEDANTA (0.71), and ICICI Bank (0.75) which indicates that the proposed model can predict true positive values of the data efficiently. Recall refers to the ability to capture all the positive instances in the model. The F1-score combines both precision and recall. From Table 2, the F1-scores for Reliance (0.80), Asian Paints (0.76), Hindustan Unilever (0.71) HDFC Bank (0.77), and VEDANTA (0.77) are very good, which shows that the proposed model can predict positive as well as negative instances effectively.

TABLE II. ACCURACY, PRECISION, RECALL AND F1-SCORE FOR SELECTED STOCKS

Name of Stock	Accuracy	Precision	Recall	F1-score
RELIANCE	0.70	0.86	0.75	0.80
KOTAK BANK	0.60	0.60	0.60	0.60
INFOSYS	0.60	0.40	0.66	0.50
ASIAN PAINTS	0.70	0.83	0.71	0.76
HINDUSTAN UNILEVER	0.60	0.83	0.63	0.71
HDFC BANK	0.70	0.83	0.71	0.77
VEDANTA	0.70	0.83	0.71	0.77
ICICI BANK	0.70	0.60	0.75	0.67

Fig. 6: Indicate the chart for testing data vs. prediction data. From the figure it is clear that the model can predict the stock's next day's closing price and the near-term trend of that stock, such as an uptrend or a downtrend. The chart represents a sample stock of RELIANCE.

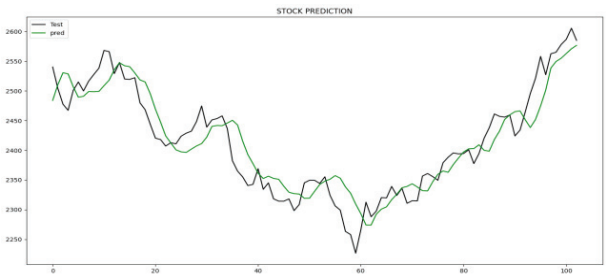

Fig. 6. Testing Data vs Predicted Data for RELIANCE Stock

V. CONCLUSION:

In conclusion, this research contributes to the growing area of literature on stock price prediction by presenting a comprehensive approach that shows the strengths of both traditional indicators and cutting-edge machine learning techniques. The performance parameters used in this study are RMSE, R2 score for price prediction, and confusion matrix for trend identification. The results showed that the precision is good in Reliance (i.e. 0.86), Asian paints (i.e. 0.83), Hindustan Unilever (i.e. 0.83.), HDFC Bank (0.83) and VEDANTA (0.83) The recall value is also good for Reliance (0.75), Asian Paints (0.71), HDFC Bank (0.71) VEDANTA (0.71) and ICICI Bank (0.75). The F1 Score achieved good values in Reliance (0.80), Asian Paints (0.76), Hindustan Unilever (0.71), HDFC Bank (0.77) and VEDANTA (0.77). The accuracy achieved for stocks such as KOTAK BANK, INFOSYS, and HINDUSTAN UNILEVER is 60%, and for RELIANCE, ASIAN PAINTS, HDFC Bank, VEDANTA and ICICI bank it is 70%. The R2 score values are $0.75 < R2 < 0.95$, which indicates a strong model. The RMSE is also reduced in all cases. The proposed methodology holds promise for financial analysts, traders, and researchers seeking more accurate and reliable predictions in the evolving environment of the stock market. This research attracts investors to invest money in the stock market.

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